Lotto & Coupon Ecosystem

Strategic Client Proposal & Technical Integration Roadmap

1. Executive Summary

This proposal outlines the seamless integration between the OzLotto / MegaDraw Gaming Platform and the CouponVault Merchant Voucher Engine. By combining a state-of-the-art online lottery system with an agile coupon infrastructure, this solution provides merchants with customer acquisition tools and offers users a frictionless way to convert gift cards into lottery ticket purchases.

2. Core Platform Capabilities

A. OzLotto / MegaDraw Platform (Gaming Backend & Portal)

- **Global Lottery Games:** Supports 8+ major formats (Powerball, MegaDraw, Saturday Lotto, Set for Life, Lucky Lotteries, Lotto Strike, Super 66).
- **Automated Draw Engine:** Cron-scheduled draws, prize tier division calculations, and instant win processing.
- **Syndicates & System Entries:** Allows group ticket purchasing and multi-number system combinations.
- **User Security & Compliance:** JWT authentication, Argon2 password hashing, 2FA enforcement, and automated KYC verification.

B. CouponVault Platform (Merchant Engine & Payment Gateway)

- **Merchant Dashboard:** Centralized hub for merchants to create, issue, track, and manage promotional gift vouchers.
- **Razorpay & Payment Gateway Integration:** Instant fiat payment processing for voucher purchases.
- **Single-Use Security Token:** Encrypted voucher redemption system guaranteeing zero double-spending.

3. Integrated User Journey & Data Flow

The integration connects merchant voucher distribution directly into the lottery wallet, driving user conversion and cross-promotions.

Step-by-Step Integration Workflow

- 1. Voucher Issuance: Merchant creates or user purchases a gift coupon (e.g., FREE50, LOTTO100) via CouponVault using Razorpay/Stripe.
- 2. Redemption Request: User navigates to the Lottery Deposit page and inputs the voucher code.
- 3. API Handshake: Fastify backend performs an authenticated API request to CouponVault to verify validity, expiry, and balance.
- 4. Wallet Credit: Upon confirmation, CouponVault locks/invalidates the voucher, and the Lottery backend instantly credits the user's wallet in Coins/USD.
- 5. Gameplay & Payout: User purchases lottery tickets for upcoming draws. Prize winnings are credited directly back to the user's wallet.

4. Technical Architecture Stack

- **Backend API:** Fastify (Node.js), TypeScript, Prisma ORM, Zod Validation.
- **Database & Cache:** PostgreSQL 16 (Relational schemas) + Redis 7 (Rate limiting, session store, queue management).
- **Coupon Engine:** Supabase, PostgreSQL Row-Level Security (RLS), Netlify/Node serverless functions.
- **Frontend Applications:** React 18, Vite, Zustand state management, TanStack React Query, Tailwind CSS.
- **Deployment:** Render (Database, Redis, Fastify Backend), Docker Compose, Netlify / Vite static hosting.

5. Project Implementation Roadmap

Phase 1: Architecture & DB Synchronization (Weeks 1 – 3)

- **Deliverables:** Setup shared API schemas, establish secure service-to-service API key authentication, and deploy database migrations across environments.

Phase 2: Payment Gateway & Coupon API Engine (Weeks 4 – 6)

- **Deliverables:** Implement Razorpay/Stripe webhooks in CouponVault, build the single-use token verification endpoint, and connect Fastify wallet handlers.

Phase 3: Frontend Integration & UX Polish (Weeks 7 – 9)

- **Deliverables:** Integrate redemption page in Lottery App, build Merchant analytics dashboard, and implement real-time toast notifications and audit logs.

Phase 4: Testing, Security Audit & Go-Live (Weeks 10 – 12)

- **Deliverables:** End-to-end integration testing, Redis rate-limit benchmarking, penetration testing, Render/Docker production deployment, and client handover.

6. Next Steps & Action Items

To initiate Phase 1, the client and development team will align on API secret management, payment gateway keys, and merchant sandbox accounts.